In [2]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

데이터셋 설명

In [3]:

```
## 데이터셋 설명
```

1. 데이터 로드

- medical_data.csv를 로드

In [13]:

```
df = pd.read_csv('medical_data.csv')
df
```

Out[13]:

	inst_id	OC	sido	openDate	bedCount	instkind	revenue	salary	
0	1	open	choongnam	20071228	175.0	nursing_hospital	4.217530e+09	2.033835e+09	2
1	3	open	gyeongnam	19970401	410.0	general_hospital	NaN	NaN	
2	4	open	gyeonggi	20161228	468.0	nursing_hospital	1.004522e+09	2.964023e+08	4
3	7	open	incheon	20000814	353.0	general_hospital	7.250734e+10	3.178605e+10	9
4	9	open	gyeongnam	20050901	196.0	general_hospital	4.904354e+10	2.446078e+10	8
...	...	...	...	...	...	...	...	...	
296	422	open	jeonnam	20050211	214.0	nursing_hospital	7.614697e+09	3.467861e+09	5
297	423	close	choongnam	20140618	NaN	hospital	6.717144e+09	0.000000e+00	-1
298	426	close	gyeongbuk	19830705	NaN	hospital	0.000000e+00	7.500000e+06	-7
299	427	close	gyeonggi	20070101	180.0	nursing_hospital	2.116892e+09	2.231944e+09	-1
300	428	open	seoul	20011017	NaN	traditional_hospital	1.340971e+09	1.318978e+08	-4

301 rows × 13 columns

2. 데이터의 처음 5개 값 출력

In [14]:

```
df.head(5)
```

Out[14]:

	inst_id	OC	sido	openDate	bedCount	instkind	revenue	salary	p
0	1	open	choongnam	20071228	175.0	nursing_hospital	4.217530e+09	2.033835e+09	225169
1	3	open	gyeongnam	19970401	410.0	general_hospital	NaN	NaN	
2	4	open	gyeonggi	20161228	468.0	nursing_hospital	1.004522e+09	2.964023e+08	41864

	inst_id	OC	sido	openDate	bedCount	instkind	revenue	salary	p
3	7	open	incheon	20000814	353.0	general_hospital	7.250734e+10	3.178605e+10	902830
4	9	open	gyeongnam	20050901	196.0	general_hospital	4.904354e+10	2.446078e+10	80749

3. 수치형 변수의 통계 정보 확인

In [15]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 301 entries, 0 to 300
Data columns (total 13 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   inst_id      301 non-null    int64
 1   OC           301 non-null    object
 2   sido         301 non-null    object
 3   openDate     301 non-null    int64
 4   bedCount     296 non-null    float64
 5   instkind     300 non-null    object
 6   revenue      293 non-null    float64
 7   salary       293 non-null    float64
 8   profit       293 non-null    float64
 9   debt         293 non-null    float64
 10  netAsset     293 non-null    float64
 11  employee     291 non-null    float64
 12  ownerChange  289 non-null    object
dtypes: float64(7), int64(2), object(4)
memory usage: 30.7+ KB
```

4. 수치형 변수 값에 결측치가 하나라도 존재하는 병원 데이터 전체를 삭제

In [16]:

```
df_ = df.dropna()
```

5. 당기순이익이 높은 상위 10개 병원의 병원 고유 번호, 영업/폐업 분류, 병원 설립일, 병원 종류 정보 출력

In [26]:

```
df_ = df_.sort_values(by='profit', ascending=False)
df_
```

Out[26]:

	inst_id	OC	sido	openDate	bedCount	instkind	revenue	salary	
292	417	open	gyeonggi	20020401	306.0	general_hospital	1.000000e+11	4.302536e+10	7.4754
276	390	open	gyeonggi	20071031	179.0	nursing_hospital	6.184769e+09	0.000000e+00	6.1847
273	384	open	ulsan	20140804	235.0	general_hospital	5.414029e+10	2.270745e+10	5.9123
272	383	open	ulsan	20160826	176.0	hospital	5.414029e+10	2.270745e+10	5.9123

	inst_id	OC	sido	openDate	bedCount	instkind	revenue	salary	
221	317	open	daegu	20070104	189.0	general_hospital	4.110397e+10	1.455766e+10	4.2654
...	...	...	...	...	...	...	...	...	
16	33	open	seoul	19820329	181.0	general_hospital	1.569070e+10	8.142441e+09	-1.9631
290	414	close	gyeonggi	20140620	200.0	general_hospital	2.375764e+09	2.668110e+09	-2.4833
15	27	open	ulsan	19810622	526.0	general_hospital	1.240000e+11	5.778814e+10	-3.7538
77	115	open	incheon	20160905	162.0	nursing_hospital	0.000000e+00	2.421379e+09	-4.6733
247	352	open	gyeonggi	19980328	523.0	general_hospital	1.410000e+11	6.403559e+10	-4.6967

279 rows × 13 columns

In [27]:

```
df_[['inst_id', 'OC', 'openDate', 'instkind']].head(10)
```

Out[27]:

	inst_id	OC	openDate	instkind
292	417	open	20020401	general_hospital
276	390	open	20071031	nursing_hospital
273	384	open	20140804	general_hospital
272	383	open	20160826	hospital
221	317	open	20070104	general_hospital
267	375	open	19920801	hospital
245	350	open	19970728	clinic
75	112	open	20070807	nursing_hospital
41	65	open	20110512	hospital
243	347	open	19920930	hospital

6. 병원 종류 별 총 직원 수의 평균값 출력

In [30]:

```
general_hospital = df_[df['instkind'] == 'general_hospital']['employee'].mean()
nursing_hospital = df_[df['instkind'] == 'nursing_hospital']['employee'].mean()

print('general_hospital 평균 값 : ', general_hospital)
print('nursing_hospital 평균 값 : ', nursing_hospital)
```

```
general_hospital 평균 값 :  384.8484848484849
nursing_hospital 평균 값 :  97.53284671532846
C:₩Users₩ixlab₩AppData₩Local₩Temp/ipykernel_21784/713454779.py:1: UserWarning: Boolean Ser
ies key will be reindexed to match DataFrame index.
  general_hospital = df_[df['instkind'] == 'general_hospital']['employee'].mean()
C:₩Users₩ixlab₩AppData₩Local₩Temp/ipykernel_21784/713454779.py:2: UserWarning: Boolean Ser
ies key will be reindexed to match DataFrame index.
  nursing_hospital = df_[df['instkind'] == 'nursing_hospital']['employee'].mean()
```

7. 대표자가 변동된 병원들의 병원 고유 번호, 영업 폐업 분류 출력

In [31]:

```
change = df_[df['ownerChange'] == 'change']

change[['inst_id', 'OC']]
```

```
C:₩Users₩ixlab₩AppData₩Local₩Temp/ipykernel_21784/4094566226.py:1: UserWarning: Boolean Se
ries key will be reindexed to match DataFrame index.
  change = df_[df['ownerChange'] == 'change']
```

Out[31]:

	inst_id	OC
177	265	close
176	264	open
228	324	open
204	296	open
21	38	open
17	34	open
287	409	open
136	202	open
96	144	open
274	385	open
72	108	open
202	291	open
35	58	close
227	323	open
286	408	open
140	208	open
139	205	open
250	355	open
219	315	open
255	360	open
284	406	open
90	137	open
158	238	close
251	356	open
275	386	open
12	22	open

	inst_id	OC
142	211	open
214	309	close
213	308	open
163	247	open
200	289	open
269	378	open
109	162	open
63	97	open
189	277	open
268	376	open
289	412	open
22	39	open
16	33	open
290	414	close
247	352	open

8. 병원 종류 별 평균 급여를 boxplot 으로 출력

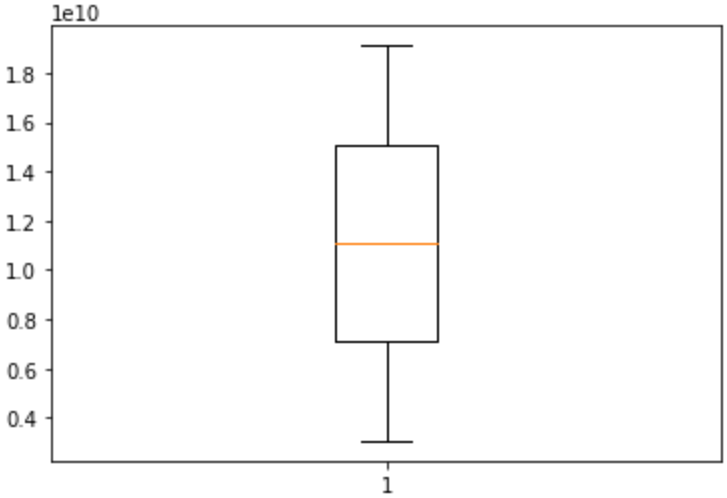
In [33]:

```
general_hospital = df_[df['instkind'] == 'general_hospital']['salary'].mean()
nursing_hospital = df_[df['instkind'] == 'nursing_hospital']['salary'].mean()

x = ['general_hospital', 'nursing_hospital']
y = [general_hospital, nursing_hospital]

plt.boxplot(y)
```

```
C:₩Users₩ixlab₩AppData₩Local₩Temp/ipykernel_21784/3945216934.py:1: UserWarning: Boolean Se
ries key will be reindexed to match DataFrame index.
  general_hospital = df_[df['instkind'] == 'general_hospital']['salary'].mean()
C:₩Users₩ixlab₩AppData₩Local₩Temp/ipykernel_21784/3945216934.py:2: UserWarning: Boolean Se
ries key will be reindexed to match DataFrame index.
  nursing_hospital = df_[df['instkind'] == 'nursing_hospital']['salary'].mean()
```

Out[33]:

```
{'whiskers': [<matplotlib.lines.Line2D at 0x29691718d00>,
  <matplotlib.lines.Line2D at 0x29691718fd0>],
 'caps': [<matplotlib.lines.Line2D at 0x296917303a0>,
  <matplotlib.lines.Line2D at 0x29691730730>],
 'boxes': [<matplotlib.lines.Line2D at 0x296917188b0>],
 'medians': [<matplotlib.lines.Line2D at 0x29691730ac0>],
 'fliers': [<matplotlib.lines.Line2D at 0x29691730e50>],
 'means': []}
```



In [ ]: